P–S–603b
9 MAY 1952
SUPERSEDING
Fed. Spec. P–S–603
20 September 1941, and
Int. Fed. Spec. P–S–603a (GSA–FSS)
1 August 1951

FEDERAL SPECIFICATION

SOAPS, POTASH-LINSEED OIL (LIQUID AND PASTE, FLOOR AND GENERAL CLEANING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers liquid- and paste-type linseed oil soaps suitable for use in washing floors and linoleum.

1.2 Classification (types). — Potash-linseed-oil soap shall be of the following types, as specified in the invitation for bids:

Type I. —Liquid
Type II.—Paste

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specifications of the issues in effect on date of invitation for bids, form a part of this specification:

P–S–536—Soap and Soap-Products (Including Synthetic Detergents); Methods of Sampling and Testing.

VV–L–791—Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling, and Testing.

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I, liquid soap.*—The soap shall be a uniform liquid soap made solely from whole neutral raw linseed oil and potash, and shall conform to the requirements set forth in this section.

3.1.2 *Type II, paste soap.*—The soap shall be a uniform translucent, firm gel or paste soap made solely from whole neutral raw linseed oil and potash, and shall conform to the requirements set forth in this section.

3.2 Composition.—The composition of the soap shall conform to the requirements shown in table I. Percentages are by weight.

3.2.1 *Computation (type II only).* — The percentage of moisture shall be computed and reported by the testing laboratory on the soap as received. The percentages of all other constituents shall be calculated and reported on an assumed moisture content of 50 percent. For basis of purchase, see 6.4.

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TABLE I.—*Composition*

	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Moisture (toluene distillation method)—percent	---	---	---	55
Total matter insoluble in alcohol—do	---	0.5	---	1.0
Free alkali, calculated as potassium hydroxide (KOH)—do	---	.05	---	.1
Free acid, calculated as oleic acid—do	---	.1	---	.2
Alkaline salts, calculated as potassium carbonate (K_2CO_3)—do	---	.1	---	.2
Matter insoluble in distilled water—do	---	.1	---	.2
Chloride, calculated as potassium chloride (KCl)—do	---	.3	---	.5
Unsaponified and unsaponifiable matter—do	---	.4	---	1.0
Anhydrous soap, calculated as potash soap—do	20	---	43	---
Total sodium compounds, calculated as Na_2O—do	---	.2	---	.5
Glycerol—do	1.8	---	4	---
Rosin	---	None	---	None
Sugar	---	None	---	None
Iodine number (WIJS) of mixed fatty acids derived from the soap	175	---	175	---
Acid number of mixed fatty acids derived from the soap	190	205	190	205

3.3 Properties.

3.3.1 *Type I, liquid soap.*—The soap shall be soluble in soft water and when diluted with water shall act as a cleaner. The flash point shall be above its boiling point (see 4.3.2). It shall not contain any solvents or oils that will damage floor surfaces. (See 6.1.) The odor shall not be objectionable in the soap as received or in a hot solution of the soap in water. The material shall not leave an objectionable odor on surfaces after washing with a water solution of the soap and rinsing thoroughly with plain water. (See 6.3.1.)

3.3.2 *Type II, paste soap.*

3.3.2.1 *Consistency, color and odor.* — The soap shall be a uniform translucent firm gel or paste soap of a yellowish-white to greenish brown color. The odor shall not be objectionable in the soap as received or in a hot solution of the soap in water. The soap shall not leave an objectionable odor on surfaces after washing with a water solution of the soap and rinsing thoroughly with plain water. (See 6.3.1.)

3.3.2.2 *Solubility and sudsing.*—The soap shall dissolve readily to give a 0.15- to 0.2-percent solution, using distilled water at 15.5° to 20° C. (60° to 68° F.). The solution so prepared shall yield at least 150 milliliters of suds when tested as prescribed in 4.3.1.2.

3.3.2.3 A solution of the soap in soft water shall act as a cleaner and shall not damage floor surfaces. (6.1.)

3.4 **Keeping qualities.**—The material of each type shall not become rancid or otherwise deteriorate when kept in a closed container.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

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4.1 **Sampling.**—Samples shall be taken as described in Federal Specifications P–S–536 and VV–L–791, as applicable.

4.2 **Inspection.**—Samples shall be inspected to determine compliance with this specification with respect to color, odor, and consistency. When required, samples taken from deliveries shall be compared with the accepted bid sample (see 6.3.1). If unsatisfactory, the soap shall be subject to rejection without further testing.

4.3 **Tests.**

4.3.1 Tests to determine conformance with the requirements of table I and 3.3.2.2 shall be performed in accordance with the applicable test methods in Federal Specification P–S–536 and as hereinafter specified.

4.3.1.1 *Moisture.*—Test shall be made on sample (type II) by method 20.2, using toluene instead of xylene.

4.3.1.2 *Sudsing test.*—Test shall be made on sample (type II) by method 200.1.

4.3.2 *Flammability* (flash and fire points).—Test shall be made on sample (type I) by method 110.3 (flash and fire points by means of open cup tester), Federal Specification VV–L–791.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Unless otherwise specified, commercial packages are acceptable under this specification.

5.2 **Packing.**—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**—Unless otherwise specified, shipping containers shall be marked with the name of the commodity, and the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

6. NOTES

6.1 **Intended use.**—Soap of each type is intended for use on all kinds of floors, except on rubber flooring.

6.2 A solution of about 1 to 2 ounces of the paste soap (type II) per gallon of soft water should be satisfactory. If the soap is not all dissolved, there will be waste and some undissolved soap may be left on the surface being cleaned. If the soap is all dissolved and the surface on which it is used is thoroughly rinsed with plain water, material conforming to type II should be satisfactory for cleaning operations. A ready-prepared liquid soap (type I) is time saving as it is in a condition to dilute and apply. If the paste soap is completely dissolved and diluted before use, it should rinse off as readily as would a ready-prepared liquid soap, since soap solutions are used in each case. If the paste soap is not completely dissolved, the use of the liquid soap would reduce waste.

6.2.1 All ordinary soaps (liquid, paste, or cake) when dissolved in a water supply containing hardness constituents will react with the latter, depositing some insoluble slimy calcium and magnesium soaps. The removal of these insoluble soaps requires thorough rinsing or mopping with plain water.

6.3 **Ordering data.** — Purchasers should specify the type of soap, and whether a bid sample is required, and should exercise any desired options offered herein. (See 1.2, 5.1, 5.2, 5.3, 6.4, and 7.2.6.)

6.3.1 *Bid samples.*—It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for

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any particular purpose, bid samples are required, invitations for bids should state that each bidder shall submit, in time to arrive prior to date of opening of bids, a sample of the soap he proposes to furnish, to show color, odor, and condition, and should specify the size and weight of sample unit required. Samples should be marked with the name of the commodity, the schedule number and date of opening of bids, and the name and address of the bidder. The accepted bid samples should be kept in air-tight containers for later comparison with samples from deliveries.

6.4 **Basis of purchase.**—Liquid soap (type I) should be purchased by volume. A gallon contains 231 cubic inches at 15.5° C. (60° F.). Paste soap (type II) should be purchased by net weight, providing the moisture does not exceed 55.0 percent. With deliveries containing 55.0 percent or less of moisture, settlement should be made on the basis of 55.0 percent of moisture; that is, 0.45 pound of nonvolatile matter should be considered 1 pound of detergent. For example: Moisture = 50.0 percent, then:

$$\text{Net weight of material to be paid for} = \frac{\text{Net weight as received} \times (100-50)}{45}$$

6.5 Federal specifications do not include all types, classes, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, and standards, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications and standards.*

Military Specification:

JAN–P–124—Packaging and Packing for Overseas Shipment — Containers (Cans, Pails, and Drums) Metal (For Other Than Subsistence Items).

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection By Attributes.

MIL–STD–129—Marking of Shipments.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 Sampling shall be conducted in accordance with Military Standard MIL–STD–105.

7.2.3 *Packing.*

7.2.3.1 *For domestic and overseas shipments.*—Approximately 50 or 100 pounds of liquid or paste soap, as specified in the contract or order, shall be packed in a round, open top, full removable bolted ring or lug cover metal pail or drum conforming to type VI of Federal Specification JAN–P–124.

[1] Applicable to Navy purchases only.

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7.2.4 *Marking.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Inspection procedures (Navy only).*—General inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

NAVY INTEREST: Md Sh S Y

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C. Prices of additional copies may be obtained from the GSA Regional Offices.)

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

Copies of this specification can be purchased for 5 cents each from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Single copies required for bidding purposes are available without charge at the nearest GSA Regional Office.

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